Market Guide for Hybrid AI Infrastructure

7 May 2025 - ID G00818310 - 13 min read

By: Dennis Smith, Chandra Mukhyala

Initiatives:I&O Platforms

Emerging AI requirements drive demand for solutions enabling AI workloads to be deployed within hybrid environments. This requires infrastructure that can handle performance demands introduced by AI workloads. This research helps I&O leaders navigate this dynamic market.

Overview

Key Findings

- AI, specifically GenAI, is transformational — particularly industry-specific models — and is becoming a top priority for all enterprises. This requires infrastructure and IT operations (I&O) teams to plan for the impact on their infrastructure strategies.
- Data is a key component of AI deployments, and since most data is still and often must stay within the confines of enterprise locations, I&O teams must prepare for infrastructure that enables the necessary processing and operationalizing of data.
- Many vendors are converging around this domain with varied compute, storage and networking offerings optimized for AI. I&O needs a structured plan to navigate the market.

Recommendations

- Make a clear identification of what use cases must have private AI infrastructure and what can be deployed on cloud provider PaaS to be certain of the scope of your use cases and capacity needs before investing.
- Differentiate between the infrastructure needed for training and fine-tuning from the infrastructure needed for inference and monitoring of deployed models as they will differ because inference can often be done on less expensive infrastructure located closer to users.
- Favor fully managed stack offerings that are conformant to industry standard reference architectures for self-managed private data center infrastructure (DCI).

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Strategic Planning Assumption

By 2028, more than 20% of enterprises will run AI workloads (training and/or inference) locally in their data centers, an increase from fewer than 2% as of early 2025.

Market Definition

Gartner defines hybrid AI infrastructure as offerings that address the need to enable AI and machine learning (ML) workloads across enterprise data centers, colocation and the edge, and public cloud. The offerings include a combination of compute, storage and networking components, along with the requisite enablement tooling, middleware and libraries. Infrastructure and operations (I&O) leaders can use these solutions to deploy AI in the most strategic location for their needs, whether on-premises, at the edge or in the cloud.

Hybrid AI infrastructure addresses enterprise needs for deployment of inference and possibly training to enterprise data centers. The solutions provide foundational capabilities for companies unable or unwilling to solely leverage public cloud AI infrastructure (e.g., due to digital sovereignty or costs involved with moving data from an existing location).

Mandatory Features

Hybrid AI infrastructure must have infrastructure and/or infrastructure-related tooling in support of the building and/or operation of traditional AI and generative AI (GenAI) workloads. This includes the enablement of:

- Automated aspects of building, deploying and maintaining AI/ML models
- Data processing and preparation of data for model training and inference
- Packaging, deploying and managing the excursion of AI/ML training and model inference, as well as fine-tuning the operation
- Integration of AI/ML libraries (e.g., PyTorch, Keras, CUDA)
- Tools enabling the deployment and operation of AI/ML workloads (e.g., InstructLab)
- Security, including encryption, access controls and compliance

Specific capabilities include the support of:

- High computational demands

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

- High-throughput, low-latency and high-capacity storage to support the large datasets and high-speed data transfers necessary to keep graphics processing units (GPUs) engaged with large quantities of data
- Low latency and lossless networking with high throughput, including near-range networks for data transfer between AI accelerators and/or CPUs (e.g., NVLink and NVLink Switch)
- High-speed, lossless networks interconnecting AI servers (e.g., InfiniBand, RDMA over Converged Ethernet [RoCE])
- The ability to support workloads on-premises, at the edge and within a public cloud hyperscaler

Common Features

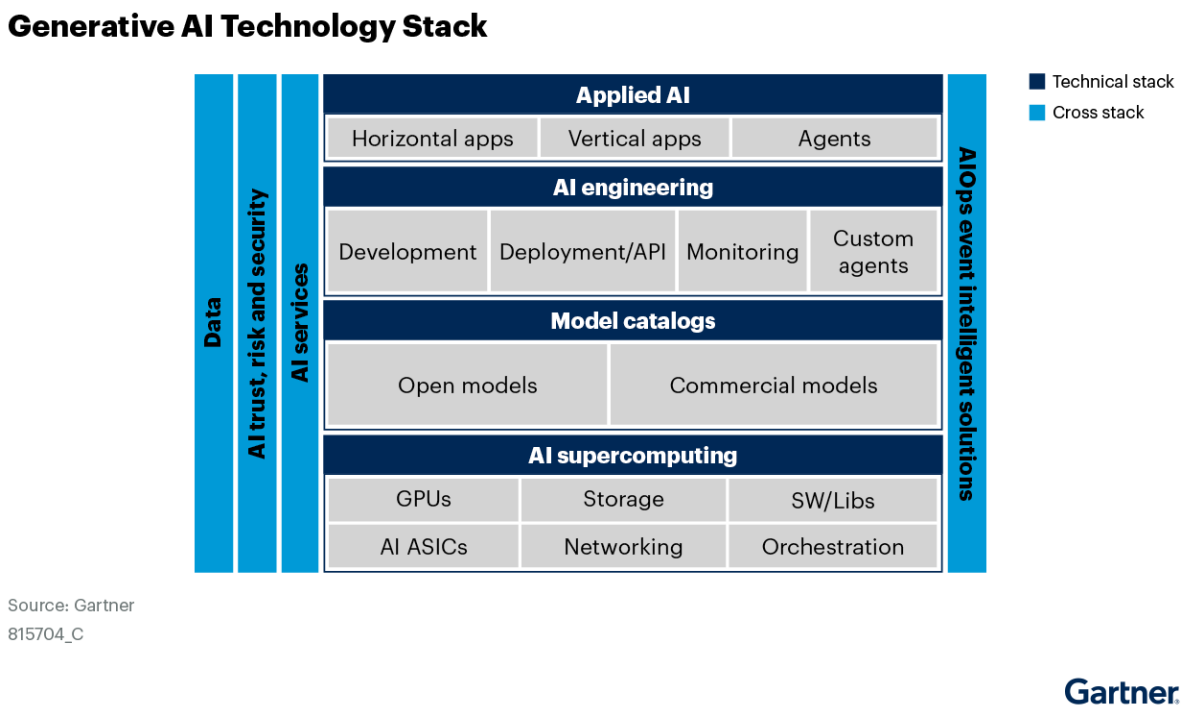
- A combination of compute, storage and networking infrastructure in support of AI workloads, which can be installed in enterprise data centers, colocation centers, at the edge and within the public cloud.
- Software that enables the operation of the underlying infrastructure that supports AI workloads. This includes some combination of automated provisioning, optimizing, observing and/or securing capabilities for the underlying infrastructure.
- Global data management that can scalably process and store high volumes of data across hybrid multicloud environments.
- Dedicated networks for training (e.g., a supercomputing network with lossless architecture) and inference (lower cost but low latency).
- Infrastructure-enabling frameworks (e.g., PyTorch, TensorFlow, JAX) and programming languages that enable workload processing.

Market Description

Enterprises have leveraged variations of AI (from traditional AI to recently released GenAI technologies) for many years, often leveraging more traditional infrastructure. GenAI and new AI use cases have placed additional demands on enterprise infrastructure due to the need to rapidly process a massive amount of data. Figure 1 depicts a GenAI technology stack.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Figure 1: GenAI Technology Stack



The bottom layer of the figure includes foundational compute, storage and networking components (AI supercomputing). While this layer shares some similarities with traditional infrastructure components such as CPU architecture and overall concepts/engineering, they are specifically designed for handling the scale and latency requirements for a GenAI workload (see How to Determine Infrastructure Requirements for On-Premises Generative AI).

The core infrastructure layer is enabled by capabilities that optimize the AI models to run on the compute hardware. This could be provided by server virtualization and management software or embedded into the hardware. Also at this layer is software that schedules the workloads to run on the infrastructure, as well as the orchestration, scheduling and provisioning of infrastructure resources. Here, Kubernetes is often deployed due to its ability to scale/contract based on workload demands, as is often seen with AI workloads.

At the next layer are the frameworks, model libraries and model deployment tools that enable data and analytics (D&A) personnel to complete their tasks (e.g., TensorFlow, PyTorch, Keras) while leveraging the models.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

The third layer consists of developer tool chains, APIs, monitoring of the agent activity and customized agents used for specific tasks. This layer also includes a data pipeline where data is infused into the process and acted on by the models deployed. Finally, the fourth layer is where the activity of the bottom three layers culminates in end-user outcomes. This could be any combination of AI-infused applications and agents often targeted for addressing industry use cases.

Besides the technology stack, there are vertical functions such as security embedded throughout that govern access to the data and encryption. Finally there is technology that allows the entire stack to be operationalized.

I&O leaders are mostly responsible for the bottom layer but can be consultative for the other layers, thus must be at least conversational on the related technology and issues (see Hype Cycle for Generative AI, 2024). I&O teams should mostly forgo procuring and integrating a wide mix of separate components that map to the elements shown in Figure 1. This requires expertise to perform these often complicated tasks. Increasingly, vendors are providing turnkey solutions (hardware and/or software) that integrate many of the components.

Market Direction

AI interest continues to rise unabated. While most current activity, particularly the training, occurs in the public cloud; increasingly there will be more of a need to bring AI to where the data is, which is currently mostly on-premises. Recent geopolitical concerns underscore this sentiment and will likely dictate on-premises as a course of action.

This means that the future of AI will have a hybrid focus (e.g., some combination of training, inference and serving spread across the public cloud and enterprise environments). Notwithstanding the operational challenges, much of the current on-premises infrastructure is not AI (specifically GenAI) workload-capable and/or optimized. AI hybrid solutions address this requirement. These solutions are provided by both traditional enterprise data center vendors, hyperscale cloud providers and emerging AI-native vendors.

Options include full-stack solutions, some provided as a service and individual AI-specialized components that must be built in-house. Most organizations will choose the former as the market is complex and rapidly evolving.

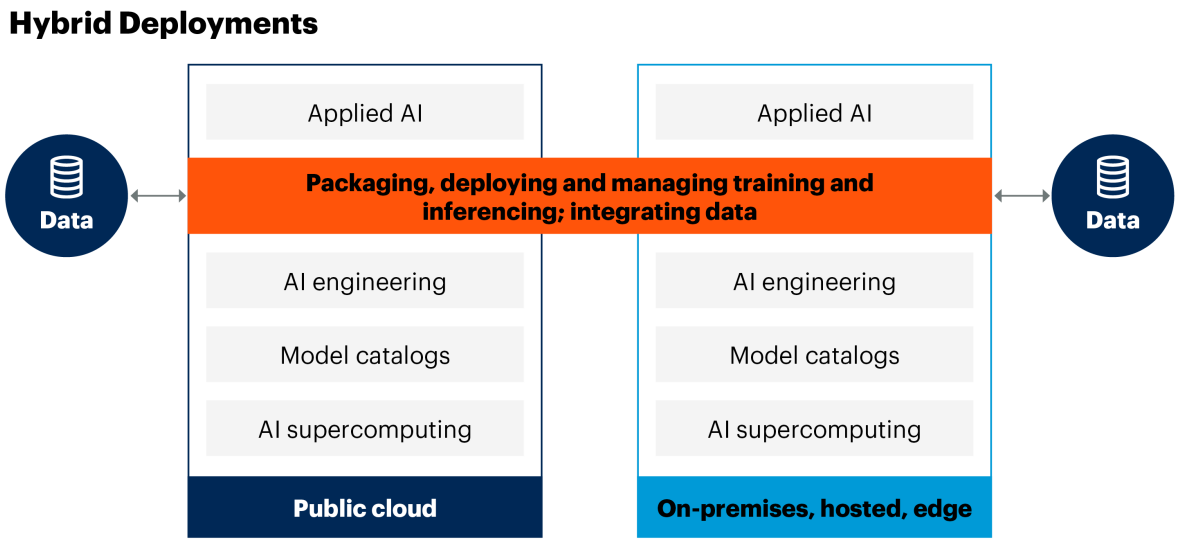
This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Market Analysis

The emphasis on supporting enterprise AI ambitions highlights the need to support AI workload on-premises, at the edge and within a public cloud hyperscaler. This need exists because some AI workloads will be deployed where the data resides due to security, compliance and performance considerations. And moving the data is often expensive.

The hybrid nature is depicted in Figure 2. Common D&A pipelines and frameworks are deployed across the infrastructure in various environments where training and inference can be combined between on-premises and public cloud environments. This makes it easier for D&A personnel to operate in multiple environments.

Figure 2: Hybrid Deployments



Source: Gartner
818310_C

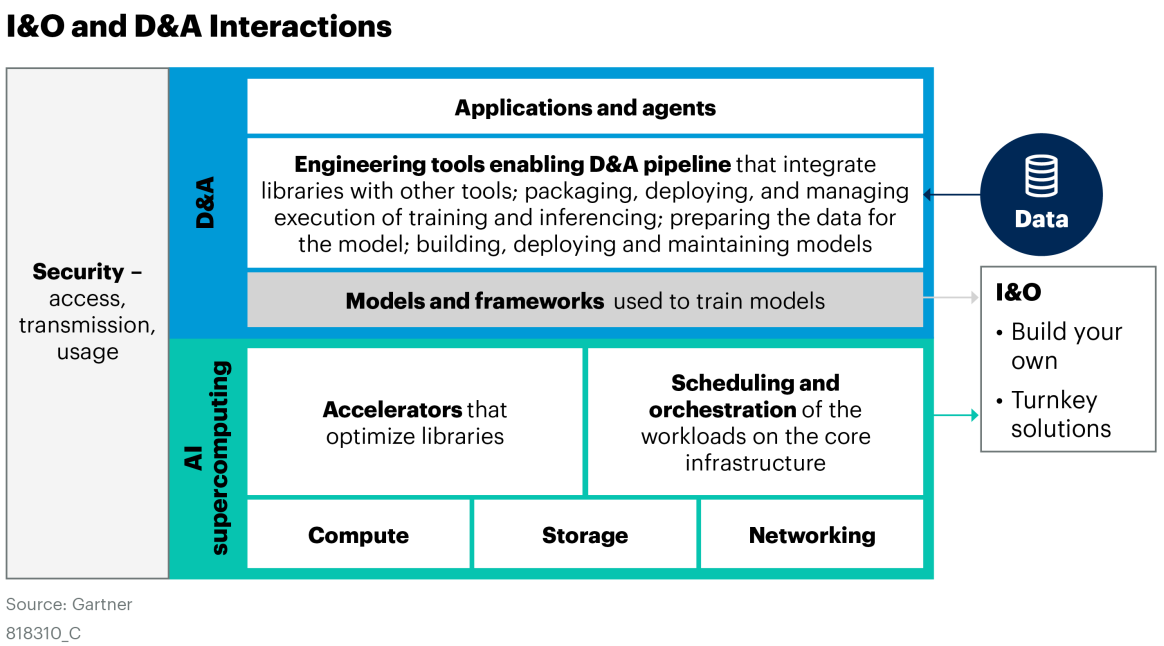
Gartner

To enable this hybrid activity, enterprises will need to deploy a combination of compute, storage and networking components that are installed in enterprise data centers, colocation, edge and within the public cloud. Additionally, software is needed that enables the operation of the underlying infrastructure that supports AI workloads. This includes some combination of automated provisioning, optimizing, observing and securing capabilities for that infrastructure.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Figure 3 shows the technology layers and interactions of I&O and D&A personnel. I&O leaders are responsible for the core infrastructure components, including their optimization and orchestration. At the very least, they must collaborate and be conversational regarding issues around the deployed models and frameworks. Increasingly, the storing and ingestion of data also requires a tight affinity with storage technologies that are operated by I&O.

Figure 3: I&O and D&A Interactions



Training Versus Inference

Hybrid AI infrastructure needs vary depending on training versus inference workloads. Training activities are associated with building the AI models, and this calls for significant resources. It requires analyzing large amounts of data to learn patterns and relationships. Inference involves using models to make predictions. It also involves applying the patterns learned during training to make predictions. This information is used to generate the ultimate results from the AI activity (e.g., generate content, make decisions or classify data).

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Retrieval-augmented generation (RAG) is a way to utilize large language models (LLMs) for natural language querying across external data sources. RAG systems augment the process of LLM inference by incorporating an external knowledge base. RAG is currently the major use case for enterprises deploying GenAI on-premises. Based on Gartner client inquiries, only a small percentage perform fine-tuning of existing LLMs on-premises and an even smaller percentage perform model training.

Hybrid AI on-premises solutions are adequate for RAG workloads and small to midsize GenAI inference. In such cases, they are also better at addressing data integration and data gravity challenges.

Core Infrastructure

Compute resources must be highly performant to meet the high computational demands. Specific to AI workloads is the requirement for specialized accelerators for processing AI data in parallel. Accelerators might include a combination of CPUs, GPUS, TPUs, FPGA and AI ASICS.

Storage must handle high throughput, low latency and high capacity to support the large datasets and high-speed data transfers necessary to keep GPUs engaged with large quantities of data. Global data management that can process and access high volumes of data across hybrid multicloud environments minimizes the need to copy data from varied locations to a separate storage. Data categorization and labeling features enable data preparation.

Storage options include all-flash NAS, object storage, distributed file systems, directly connected GPU storage and parallel file systems. Performance acceleration features like NVIDIA GPUDirect and NFS over RDMA are highly desirable.

Networking requires low latency and lossless networking with high throughput, including: near-range networks for data transfer between AI accelerators and/or CPUs (e.g., NVIDIA NVLink and NVIDIA NVLink Switch). Additional requirements include high-speed, lossless networks interconnecting AI servers (e.g., InfiniBand, RoCE). In some instances, an enterprise should consider dedicated networks for training (e.g., supercomputing network with lossless architecture) and inferencing (lower cost but low latency). The storage and networking must keep the compute resources highly utilized.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Optimizers, Orchestrators, Schedulers and Frameworks

The size of the models used requires it to run efficiently on hardware. Optimizers perform this function (e.g., OctoAI acquired by NVIDIA in September 2024; DNN compilers). I&O must collaborate with their D&A colleagues in the selection and use of these capabilities. Orchestration and scheduling of the underlying infrastructure is also required. This is often performed by a Kubernetes deployment which is most often managed and operated by I&O.

Frameworks (and their associated libraries) enable D&A personnel to more quickly build, deploy and operate AI workloads. Among these are open-source (e.g., TensorFlow, PyTorch, Keras) and proprietary offerings (Microsoft Cognitive Toolkit, H20.ai and Amazon Web Services Apache MXNet). There is also an integration of AI/ML libraries (e.g., PyTorch, Keras, NVIDIA CUDA, etc.) that is often used.

Infrastructure-enabling frameworks (e.g., PyTorch, TensorFlow, JAX) and programming languages that enable workload processing might also be used, along with tools enabling the deployment and operation of AI/ML workloads (e.g., Red Hat InstructLab). I&O must familiarize itself with the offerings that will be used by D&A personnel in your organization.

D&A personnel leverage automated aspects of packaging, deploying and managing the excursion of AI/ML training and model inference, as well as fine-tuning. Additionally, they perform data processing and preparation of data for model training, inference and serving. This is the top layer of the technology stack that is dependent on each lower layer. From this technology stack, applications (AI agents) are enabled.

Since data is the key input for all AI activity, security is key. This includes encryption, access controls and compliance. I&O will need to engage with all entities in your organization that handle security matters.

Representative Vendors

The vendors listed in this Market Guide do not imply an exhaustive list. This section is intended to provide more understanding of the market and its offerings.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Vendor Selection

In this research, we have included a sampling of vendors and offerings (see Note 1). We have excluded hyperscalers; for those vendors, please see Magic Quadrant for Strategic Cloud Platform Services, Critical Capabilities for Strategic Cloud Platform Services, Magic Quadrant for Distributed Hybrid Infrastructure and Critical Capabilities for Distributed Hybrid Infrastructure.

The vendors mentioned in this section were selected to represent diverse Hybrid AI infrastructure solutions. We selected a representative sample that has garnered reasonable interest from among Gartner clients and represents the range of capabilities typically offered by providers in the market.

In all cases, Gartner confirmed through public sources that the vendor offers a solution that is generally available. Note that this compilation is not intended to be an exhaustive, complete list, nor does it rank providers or predict which providers/solutions are most likely to succeed as the market evolves.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Table 1: Representative Vendors in AI Hybrid Infrastructure

(Enlarged table in Appendix)

Vendor	Solutions
AMD	AMD Instinct MI300 Series accelerators and AMD Pensando Networking
Arista Networks	Arista Etherlink
Broadcom	VMware Private AI Foundation with NVIDIA
Cerebras	CS-2 and CS-3 compute platforms
Cisco	Unified Computing System (UCS) and Cisco Intersight Infrastructure Service
Cohesity	Cohesity Gaia
DDN	AI400X2 Turbo and DDN Infinia
Dell Technologies	Dell AI solutions
Groq	Compute clusters
Hitachi Vantara	Hitachi iQ
Hewlett Packard Enterprise (HPE)	HPE Private Cloud AI
Huawei	Atlas 900 PoD, Atlas 900 AI Cluster
IBM (Red Hat)	Red Hat AI
Intel	Intel AI accelerators and acceleration engines; also GPUs and NPUs
Juniper	AI Data Center Networking
Lenovo	Lenovo for Hybrid AI
NetApp	NetApp AIPod
Nutanix	Nutanix GPT-in-a-Box
NVIDIA	NVIDIA Hopper and Blackwell Architectures
Pure Storage	AI-Ready Infrastructure (AIRI)
Qualcomm	AI100 and AI100 Ultra
SambaNova	SambaNova Suite and DataScale
Scale Computing	Scale Computing Platform
SoftBank Group (Graphcore)	Hardware accelerators
Supermicro	Generative AI SuperCluster
SUSE	SUSE AI
VAST	VAST Data Platform
WEKA	WEKA Data Platform

Source: Gartner (May 2025)

Market Recommendations

I&O leaders considering hybrid AI infrastructure solutions:

- Heavily engage with D&A personnel working on AI projects within your organization as you will need to collaborate on certain aspects of the technology stack.
- Approach AI strategy with a hybrid mindset, ensuring that both the cloud and data center aspects are in coordination.
- Connect with current strategic technology vendors on their AI offerings and roadmaps; these evolve rapidly and it is important to stay up to date.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

- Avoid trying to build your own versus a turnkey offering, unless you have the resources required to execute.

Note 1: Gartner's Initial Market Coverage

This Market Guide provides Gartner's initial coverage of the market and focuses on the market definition, rationale for the market and market dynamics.

Recommended by the Authors

Some documents may not be available as part of your current Gartner subscription.

Innovation Guide for Generative AI Technologies

Lessons From Generative AI Early Adopters

© 2025 Gartner, Inc. and/or its affiliates. All rights reserved. Gartner is a registered trademark of Gartner, Inc. and its affiliates. This publication may not be reproduced or distributed in any form without Gartner's prior written permission. It consists of the opinions of Gartner's research organization, which should not be construed as statements of fact. While the information contained in this publication has been obtained from sources believed to be reliable, Gartner disclaims all warranties as to the accuracy, completeness or adequacy of such information. Although Gartner research may address legal and financial issues, Gartner does not provide legal or investment advice and its research should not be construed or used as such. Your access and use of this publication are governed by Gartner's Usage Policy. Gartner prides itself on its reputation for independence and objectivity. Its research is produced independently by its research organization without input or influence from any third party. For further information, see "Guiding Principles on Independence and Objectivity." Gartner research may not be used as input into or for the training or development of generative artificial intelligence, machine learning, algorithms, software, or related technologies.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Table 1: Representative Vendors in AI Hybrid Infrastructure

Vendor ↓	*Solutions* ↓
AMD	AMD Instinct MI300 Series accelerators and AMD Pensando Networking
Arista Networks	Arista Etherlink
Broadcom	VMware Private AI Foundation with NVIDIA
Cerebras	CS-2 and CS-3 compute platforms
Cisco	Unified Computing System (UCS) and Cisco Intersight Infrastructure Service
Cohesity	Cohesity Gaia
DDN	AI400X2 Turbo and DDN Infinia
Dell Technologies	Dell AI solutions
Groq	Compute clusters
Hitachi Vantara	Hitachi iQ
Hewlett Packard Enterprise (HPE)	HPE Private Cloud AI
Huawei	Atlas 900 PoD, Atlas 900 AI Cluster
IBM (Red Hat)	Red Hat AI
Intel	Intel AI accelerators and acceleration engines; also GPUs and NPUs
Juniper	AI Data Center Networking

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Vendor ↓	Solutions ↓
Lenovo	Lenovo for Hybrid AI
NetApp	NetApp AIPod
Nutanix	Nutanix GPT-in-a-Box
NVIDIA	NVIDIA Hopper and Blackwell Architectures
Pure Storage	AI-Ready Infrastructure (AIRI)
Qualcomm	AI100 and AI100 Ultra
SambaNova	SambaNova Suite and DataScale
Scale Computing	Scale Computing Platform
SoftBank Group (Graphcore)	Hardware accelerators
Supermicro	Generative AI SuperCluster
SUSE	SUSE AI
VAST	VAST Data Platform
WEKA	WEKA Data Platform

Source: Gartner (May 2025)

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.